



AZ-900 Module 1: Cloud Concepts

(COMPLETE EXAM-READY REVISION)



1. What is Cloud Computing?

Definition

Cloud computing = **Delivering computing services over the internet** (“the cloud”) using a **pay-as-you-go** pricing model.

Key Benefits

- **Cost-efficient** (no upfront CapEx)
 - **Scalable** (scale out/in)
 - **Reliable** (SLAs)
 - **Global reach** (datacenters worldwide)
 - **Security** (shared responsibility)
 - **Agility & speed** (deploy in minutes)
 - **High availability** (uptime)
 - **Elasticity** (auto-scale)
 - **Disaster recovery options**
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2. Cloud Types

Public Cloud

- Owned by cloud provider (Azure)
- Shared hardware
- Most common
- Example: Azure, AWS

Private Cloud

- Dedicated hardware for one company
- On-prem or hosted
- More control, more cost

Hybrid Cloud

- **Mix** of public + private
- Most flexible
- Used when:
 - Compliance needed
 - Sensitive data on-prem
 - Gradual cloud migration

Multi-Cloud

- Using **multiple providers** (Azure + AWS + GCP)
 - Avoid vendor lock-in
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3. Cloud Service Models

IaaS (Infrastructure as a Service)

You manage: **OS, apps, data**

Provider manages: **servers, networking, virtualization**

Examples:

- Azure VM
- Azure Storage
- VNets

Think: “**Renting hardware**”

PaaS (Platform as a Service)

You manage: **Apps + data**

Provider manages: **OS, hardware, runtime**

Examples:

- Azure App Service
- Azure SQL Database

Think: “**Just deploy code**”

SaaS (Software as a Service)

Provider manages **everything**.

Examples:

- Microsoft 365
- Outlook.com
- Azure DevOps (some parts)

Think: “**Just use the application**”

4. Cloud Characteristics (MUST REMEMBER!)

On-demand self-service

Provision resources instantly.

Pay-as-you-go

You pay for what you consume.

Elasticity

Resources adjust automatically.

Scalability

Vertical: Bigger machine

Horizontal: More machines (preferred)

High Availability

99.9% uptime guaranteed by SLAs.

Fault Tolerance

Service continues even if components fail.

Global Reach

Deploy anywhere.

Agility

Move fast, experiment quickly.

5. Shared Responsibility Model

Area	Microsoft	You
Physical security	✓	✗
Network	✓	✗
Datacenter	✓	✗
OS (IaaS)	✗	✓
Apps	✗	✓
Identity & access (Azure AD)	Shared	Shared
Data	✗	✓ (Always your responsibility)



6. CapEx vs OpEx

CapEx

- Upfront costs
- Hardware purchase
- On-prem

OpEx

- Pay-as-you-go
- Subscription-based
- Cloud model

👉 Exam Trick:

Cloud = **OpEx**,
On-prem = **CapEx**

7. Cloud Cost Benefits

- No upfront hardware cost
 - No maintenance cost
 - No datacenter management
 - Autoscaling = cost control
 - Pay only for what you use
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8. Azure Architecture

Regions

- Physical location (ex: East US)
- Contains datacenters

Availability Zones

- **Physically separate locations** within a region
- Protect from datacenter-level failure
- Master for **high availability**

Region Pairs

- Each region is paired with another (ex: East US ↔ West US)
 - Used for disaster recovery
 - Updates rolled out pair-wise
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9. Services for Availability & Resiliency

Availability Set

Protects against:

- Hardware failure
- Rack failure

Components:

- **Update Domain (UD)**
- **Fault Domain (FD)**

Availability Zones

Protects against:

- Entire datacenter outage

More resilience than Availability Sets.



10. Vertical vs Horizontal Scaling

Vertical Scaling

Increase VM size
(Scale up/down)

Horizontal Scaling

Add more VMs
(Scale out/in)

👉 Cloud prefers **Horizontal scaling**.



11. Azure Management Concepts

Resources

Basic unit (VM, DB, storage)

Resource Groups (RG)

Container for resources

RG = **logical grouping**

Subscriptions

Billing boundary

Management Groups

Grouping of subscriptions

Used for applying policies across org



12. SLA — Service Level Agreement

- Defines **uptime guarantee**
- Example: 99.9%, 99.95%, 99.99%

Higher SLA = combine resources:

- Load balancer + 2 VMs = higher SLA
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13. Defense in Depth (Security Layers)

Layers (**Memorization technique: PIP-N-CAD**) :

1. Data

2. Application
 3. Compute
 4. Networking
 5. Perimeter
 6. Identity & Access
 7. Physical
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14. Serverless Computing

You manage: **Just the code**

Azure manages: **Everything else**

Examples:

- Azure Functions
- Azure Logic Apps

Billing:

Based on **executions**, not uptime.



FLASHCARDS (QUICK REVISION)

Flashcard 1

Q: Public Cloud?

A: Owned by provider, shared hardware, pay-as-you-go.

Flashcard 2

Q: Private Cloud?

A: Dedicated to one organization.

Flashcard 3

Q: Hybrid Cloud?

A: Combination of public + private.

Flashcard 4

Q: IaaS example?

A: Virtual Machine.

Flashcard 5

Q: PaaS example?

A: Azure App Service / SQL Database.

Flashcard 6

Q: SaaS example?

A: Microsoft 365.

Flashcard 7

Q: CapEx vs OpEx?

A: CapEx = upfront investment; OpEx = pay-as-you-go.

Flashcard 8

Q: Scalability vs Elasticity?

A: Scalability = manual scaling; Elasticity = auto-scaling.

Flashcard 9

Q: Availability Zones?

A: Physically separate datacenters within a region.

Flashcard 10

Q: Shared Responsibility — Data?

A: Customer's responsibility.

Flashcard 11

Q: Serverless computing?

A: Run code without managing servers.

Flashcard 12

Q: Region Pair purpose?

A: Disaster recovery & sequential updates.

Flashcard 13

Q: Consumption-based model?

A: Pay for what you use.

AZ-900 Module 2 — Describe Azure Architecture & Services

(COMPLETE EXAM-READY REVISION)

1. Azure Architecture Components

Azure Regions

- Physical locations around the world
- Example: East US, West Europe, Central India
- Services availability varies by region

Important:

Choose the region **closest to users** for lowest latency.

Availability Zones (AZs)

- Physically separate datacenters *within* a region
- Each zone has independent:
 - Power
 - Cooling
 - Networking

Used for:

- ✓ VM redundancy
- ✓ Load-balanced apps
- ✓ Critical databases
- ✓ Storage (ZRS)

Provides: Datacenter-level fault tolerance

Region Pairs

- Linked regions in same geography
- Benefits:
 - Disaster recovery
 - Sequential updates
 - Data residency requirements

Example: **East US** ↔ **West US**

Resource Groups (RGs)

- Logical containers to store Azure resources
- Deleting RG deletes all resources inside
- Good for grouping by:
 - App
 - Department
 - Environment (Dev/Test/Prod)

Subscriptions

- **Billing boundary**
- Controls spending
- Assign policies + RBAC at subscription level
- One subscription can contain many RGs

Management Groups

- Used to organize **multiple subscriptions**
- Great for large organizations
- Support applying Azure Policy across many subscriptions

Hierarchy to remember:

Management Group → Subscription → Resource Group → Resources



2. Azure Compute Services



Azure Virtual Machines (IaaS)

- Complete control over OS
- Choose CPU, RAM, size
- Supports Linux & Windows
- Lift-and-shift migration



Virtual Machine Scale Sets

- Automatically scale VMs based on demand
- Used for large, distributed applications



Azure App Service (PaaS)

Host:

- Web Apps
- APIs
- Mobile Apps

Features:

- Auto-scaling
- Built-in CI/CD
- Supports multiple languages (.NET, Java, Python, Node)

Azure Functions (Serverless)

- Event-driven
- Triggered by:
 - HTTP request
 - Timer
 - Queue message
 - Blob upload
- Pay per execution

Key Point:

You only upload code; Azure manages the rest.

Azure Kubernetes Service (AKS)

- Managed container orchestration
 - Auto-scaling
 - Helps deploy microservices
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3. Azure Networking Services

Virtual Network (VNet)

- Private network in Azure
- Allows VMs and services to communicate securely
- Supports:
 - Subnets
 - Network Security Groups (NSGs)

- Routing

VPN Gateway

- Secure encrypted connection (IPsec)
- Connects:
 - On-prem → Azure
 - Azure → Azure

ExpressRoute

- Private fiber connection
- Not over the public internet
- High reliability & speed
- Used by enterprises

Load Balancer

- Layer 4 (TCP/UDP) load balancing
- Used for:
 - VMs
 - Internal or external balancing

Application Gateway

- Layer 7 (HTTP/HTTPS)
- Supports:
 - Web Application Firewall (WAF)
 - URL-based routing
 - SSL termination



Azure Front Door

- Global load balancing
 - CDN + caching
 - High performance for global apps
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4. Azure Storage Services



Blob Storage

- Object storage
- Stores:
 - Images
 - Videos
 - PDFs
 - Backups
- Access tiers:
 - Hot → frequent use
 - Cool → infrequent use
 - Archive → rarely used data



Azure File Storage

- Managed SMB file shares
- Lift-and-shift replacement for file servers
- Accessible by both Azure & On-prem machines

Azure Disk Storage

- Disks for VMs
- Types:
 - Premium SSD
 - Standard SSD
 - HDD

Queue Storage

- Message queuing
- Used for decoupling applications

Table Storage

- NoSQL key-value storage
- Cheap and scalable

Storage Redundancy Options

MOST EXAM-TESTED
Know these VERY well.

Redundancy	Copies	Location
LRS	3 copies	Same datacenter
ZRS	3 copies	Across zones
GRS	LRS + copy	Secondary region
GZRS	ZRS + copy	Secondary region

5. Azure Database Services

Azure SQL Database (PaaS)

- Automatic patches
- High availability
- Serverless compute option
- No OS management

Azure SQL Managed Instance

- More compatibility with SQL Server
- Supports SQL Agent
- Good for migrating on-prem SQL

Azure Cosmos DB

- NoSQL database
- Multi-model:
 - Document (Mongo)
 - Key-value
 - Column
 - Graph
- Global distribution
- 99.999% SLA
- Millisecond read latency

Azure Databases for open-source:

- MySQL

- PostgreSQL
- MariaDB

Fully managed PaaS services.

6. Web, Mobile & IoT Services

Azure Notification Hubs

- Push notifications to mobile devices

API Management

- Manage all APIs in one place
- Features:
 - Rate limiting
 - Versioning
 - Authorization
 - Developer portal

IoT Hub

- Secure communication with IoT devices
- Central messaging hub

IoT Central

- Fully managed SaaS IoT solution
- Easy, no infrastructure to manage



7. AI & Machine Learning Services



Azure Cognitive Services

APIs for:

- Vision
- Speech
- Language
- Decision
- Search



Azure Machine Learning

- Build, deploy, train models
- Automated ML
- Designer UI



8. Azure Integration Services



Logic Apps

- Visual workflow automation
- Integrates with:
 - Office 365

- Salesforce
- SQL
- SAP



Service Bus

- Enterprise messaging
- Used for reliable message delivery



Event Grid

- Event routing service
- Serverless
- Supports nearly any Azure service



Event Hubs

- Real-time event streaming
- Millions of events per second



9. Azure Management Tools



Azure Portal

- Web UI for managing resources



PowerShell

- Command-based automation

- Uses Azure PowerShell module



Azure CLI

- Cross-platform command-line tool
- Uses **az** commands



ARM Templates

- JSON Infrastructure-as-Code
- Declarative
- Repeatable deployments



Azure Advisor

- Recommendations for:
 - Cost
 - Security
 - Reliability
 - Performance
 - Operational excellence



Azure Monitor

- Collects metrics + logs
- Creates dashboards
- Alerting + diagnostics



Azure Service Health

- Health of *Azure services* globally

- Outages + planned maintenance

Azure Resource Health

- Health of *your specific resources*
 - States: Available / Degraded / Unavailable
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FLASHCARDS (QUICK REVISION)

Flashcard 1

Q: Region vs Availability Zone?

A: Region = geographic area; AZ = separate datacenters in a region.

Flashcard 2

Q: App Service type?

A: PaaS for web apps.

Flashcard 3

Q: Azure Functions?

A: Serverless, event-driven compute.

Flashcard 4

Q: Blob storage used for?

A: Unstructured data (images, video, backup).

Flashcard 5

Q: VNet?

A: Private virtual network in Azure.

Flashcard 6

Q: LRS vs ZRS?

A: LRS = one datacenter; ZRS = multiple zones.

Flashcard 7

Q: Cosmos DB?

A: Global, multi-model NoSQL DB.

Flashcard 8

Q: ExpressRoute?

A: Private dedicated connection (not internet).

Flashcard 9

Q: Service Bus?

A: Enterprise messaging (reliable).

Flashcard 10

Q: Event Grid?

A: Event routing for serverless workflows.

AZ-900 Module 3 — Describe Azure Management & Governance

(COMPLETE, EXAM-READY SUMMARY)

1. Azure Identity, Access, and Security

Azure Active Directory (Azure AD / Entra ID)

Microsoft's cloud-based identity & access management.

Key Features:

- Authentication (sign-in)
- Authorization (access to resources)
- SSO (Single Sign-On)
- Conditional Access
- MFA (Multi-Factor Authentication)
- Identity Protection

Azure AD is NOT:

- A domain controller
- AD DS (traditional Active Directory)

Azure AD Users & Groups

- Used for assigning permissions

- *Security groups* → assign RBAC roles
- *Microsoft 365 groups* → collaboration

Role-Based Access Control (RBAC)

Controls **who** can do **what** on **which resource**.

Three key components:

1. **Security Principal** → User / Group / App
2. **Role Definition** → Reader, Contributor, Owner
3. **Scope** → Management Group / Subscription / RG / Resource

Built-in Roles:

- **Owner** → full control
 - **Contributor** → no access management
 - **Reader** → view only
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2. Azure Security Services



Azure Key Vault

- Securely stores:
 - Secrets
 - Keys
 - Certificates
- Prevents hardcoding credentials
- Centralized key management

Microsoft Defender for Cloud

Provides:

- Threat protection
- Security posture management (Secure Score)
- Recommendations
- Alerts

Two components:

- **CSPM** → Posture management
 - **CWP** → Workload protection
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3. Azure Governance Tools

Azure Policy

- Enforces organizational rules
- Ensures resources are compliant

Examples:

- Allowed VM sizes
- Enforce tags
- Require encryption
- Prevent Public IPs

Outputs:

- **Compliant / Non-Compliant**

Azure Blueprint

Package that includes:

- Policies
- Role assignments
- ARM templates
- Resource groups

Used to deploy **standardized environments** across the organization.

Think:

Blueprint = Policy + RBAC + ARM Template bundle.

Azure Resource Manager (ARM)

- Deployment & management engine
- Enables RBAC, tags, policies
- Uses **ARM Templates**

ARM Templates

- **JSON Infrastructure-as-Code**
- Declarative (state-based)
- Repeatable deployments
- Version-controlled
- Shared across teams

Tags

- Key-value pairs
- Used for:
 - Cost management

- Organization
- Automation

Examples:

Environment = Production

Department = Finance



4. Azure Management Tools



Azure Portal

- Web-based GUI
- Easiest way to manage Azure resources



Azure PowerShell

- Command-based automation
- Uses PowerShell `Az` module



Azure CLI

- Cross-platform
- Developer-friendly
- Uses `az` commands



Azure Cloud Shell

- Browser-based shell
- Supports: PowerShell & Bash
- Pre-installed tools



5. Monitoring & Management Services



Azure Monitor

Collects:

- Metrics
- Logs
- Alerts
- Dashboards

Used for:

- Performance monitoring
- Diagnostics
- Troubleshooting



Log Analytics Workspace

Stores logs from:

- Azure Monitor
- VMs
- Apps
- Networks

Uses **Kusto Query Language (KQL)**.



Azure Service Health

Shows:

- Azure outages
- Service incidents
- Planned maintenance

This is about **Azure's health**, not your resources.

Azure Resource Health

Shows:

- Health of **your individual resources**
- States:
 - Available
 - Degraded
 - Unavailable

Gives troubleshooting guidance.

6. Azure Cost Management & Billing

Cost Management

Tracks:

- Cost analysis
- Forecasting
- Budgets
- Alerts
- Recommendations (from Advisor)



Budgets

- Set spending limits
- Trigger alerts at threshold (e.g., 80%)



Azure Advisor (Cost Recommendations)

Can recommend:

- Shutdown unused VMs
- Resize VMs
- Buy reserved instances
- Remove idle Public IPs
- Use cost-effective SKU options



Azure Reservations (Reserved Instances)

- Commit for **1 year or 3 years**
- Up to **72% savings** for:
 - VMs
 - SQL DB
 - Cosmos DB
- Best for predictable workloads



Azure Hybrid Benefit

- Reuse **Windows Server / SQL Server** licenses
- Saves 40%+ on cost



Total Cost of Ownership (TCO) Calculator

Compares:

- On-prem costs
vs
- Azure costs

Helps justify cloud migration.

FLASHCARDS (SUPER FAST REVISION)

Flashcard 1

Q: Azure AD purpose?

A: Identity and access management.

Flashcard 2

Q: RBAC vs Policy?

A: RBAC = *who can do what*; Policy = *what resources must comply with*.

Flashcard 3

Q: Key Vault used for?

A: Storing secrets, keys, certs securely.

Flashcard 4

Q: Azure Advisor categories?

A: Cost, Security, Reliability, Performance, Operational Excellence.

Flashcard 5

Q: Service Health vs Resource Health?

A: Service Health = Azure-wide; Resource Health = Your resource.

Flashcard 6

Q: Azure Policy?

A: Enforces organizational rules & compliance.

Flashcard 7

Q: Blueprint?

A: Bundle of policies + RBAC + ARM templates.

Flashcard 8

Q: Reservations?

A: 1/3-year commitment for major savings.

Flashcard 9

Q: TCO Calculator?

A: Estimate cost savings when moving to Azure.

Flashcard 10

Q: ARM Template?

A: JSON-based Infrastructure-as-Code.
